Exam 1 Corrections

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I had to take pictures to keep my exam eligible for UEC credit, and those pictures were not very good so I have to rewrite them. It is faster and more convenient for me to type these into a LaTeX document and upload them than to rewrite everything and scan them in (plus it creates a document that's easier to keep track of and refer back to in my files!) so I will do that here. Hopefully that's fine.

1 Multiple Choice Questions

These appear to all be right (unless I am missing something?) No corrections to be done.

2 Computation Question 1

Part A was entirely correct. On Part B I simply forgot what K-nearest neighbors regression is – the idea is that in K-nearest neighbors *regression*, you identify predictions for point values based on their numerical closeness to others in the dataset. This requires that the points be quantitative, which this dataset is not because the y-value is a choice between Cherry and White. (The dataset could presumably be transformed into an appropriate one by turning y into an indicator variable, but this is still a transformation fo the existing dataset.)

3 Computation Question 2

I lost half a point on this question because I did not specify that R^2 measures the amount of variation captured in the response variable by a *linear* relationship with the predictor variable. I assumed that would be obvious since we are doing linear regression, but I

should have been more clear here.

4 Computation Question 3

We are asked to calculate the residual standard error $s = \sqrt{15}$. The marking on my test says I did the residual standard error right, so I will simply copy this step:

x	y	$\hat{y}$	$e_i = y - \hat{y}$
2	5	10	-5
5	21	19	2
7	30	25	5
10	35	34	1
14	44	46	-2
17	54	55	-1

The sum of the standard errors is $SSE = -5^2 + 2^2 + 5^2 + 1^2 + -2^2 + -1^2 = 25 + 4 + 25 + 1 + 4 + 1 = 50 + 8 = 60$. There are 6 data points and 2 degrees of freedom are used, so we get the MSE by computing $\frac{60}{6-2} = 15$, so the RSE is $\sqrt{15}$, which is what we want. My problem here was that I forgot that I had to increase the number of predictors by 1 to get the degrees of freedom. I am honestly not sure why I did that; I am fine with the concept.

5 Computation Question 4

My picture is blurry here, but it looks like we are given $\sum x_i = 18$, $\sum x_i^2 = 70$, and $SSE = \sum (y_i - \hat{y}_i)^2 = 24$. It also appears we are summing six values, which means we have $n = 6$ data points. We want the estimated variance of $\hat{\beta}_1$ (I think; my picture is not great!) which should just be the square of the standard error of b_1. If we do this problem right we should get $se(b1) = \frac{3}{8}$ here.

We have $SSE = 24$, so we can calculate $MSE = \frac{24}{6-2} = 6 = s^2$. We know that we can write $se(b_1) = s\sqrt{\frac{1}{\sum (x_i - \bar{x})}}$, which we can rearrange to the variance by writing

$$se^2(b_1) = \frac{s^2}{\sum x_i^2 - n\bar{x}^2} = \frac{6}{70 - 6 \times \left(\frac{18}{6}\right)^2} = \frac{6}{70 - 18 \times 3} = \frac{6}{16} = \frac{3}{8}$$

which is what we want! I did two things wrong here: first, I repeated the same mistake using $n - 1$ instead of $n - 2$ for the degrees of freedom, and second, I wrote down $\sum x_i^2 -$

$n\bar{x}$ for the variance of x instead of $\sum x_i^2 - n\bar{x}^2$.

6 Computation Question 5

We are given here $n = 10$, $se^2(x) = 100$, $SST = 300$, and $SSReg = 100$; this immediately means that $SSE = 200$. We need to calculate the F-statistic, which really means we need to calculate the t-statistic since $F = t^2$. We should get $F = 4$ in the end.

According to the study guide, we can actually shortcut this since it's simple linear regression:

$$F = \frac{SSReg}{MSE} = \frac{100}{\frac{200}{10-2}} = \frac{100}{25} = 4$$

However, let's solve this as $F = t^2$ since that was the approach I used on the test. This means we need to find $t = \frac{b_1}{se(b_1)}$. I didn't remember the appropriate formula for b_1 here, and I even had a question mark written on the test on the part I was unsure about! Looking back, what I should have done for my approach was use

$$SSReg = b_1^2 \sum (x_i - \bar{x})^2 \implies 100 = b_1^2 \times 100 \implies b_1^2 = 1$$

So as usual, this problem just devolves to finding the standard error of b_1. We found before that $MSE = \frac{200}{10-2} = 25$, so we get $se^2(b_1) = \frac{25}{100} = \frac{1}{4}$ and thus

$$F = \frac{b_1^2}{se^2(b_1)} = \frac{1}{\frac{1}{4}} = 4$$

as required.